Bharath Ram
Data Engineer

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Summary:

Senior Data Engineer with 9+ years building enterprise-scale ETL/ELT pipelines across Azure (Data Factory, Databricks, Synapse) and AWS (Glue, EMR, Redshift) platforms, processing terabytes of transactional data daily. Deep expertise in PySpark, Spark Structured Streaming, Delta Lake, Apache Airflow orchestration, and dimensional modeling. Led full-cycle Lakehouse implementations delivering measurable improvements in data quality, performance, and operational efficiency.

Technical Summary

- Extensive experience with Azure cloud services including Azure Data Factory, Databricks, Synapse Analytics, Data Lake Storage Gen2, Event Hubs, Functions, Monitor, DevOps, Key Vault, and Cosmos DB.
- Proficient in AWS cloud ecosystem including Glue, EMR, Redshift, S3, Lambda, Step Functions, Kinesis, Athena, CloudWatch, CloudFormation, and DynamoDB.
- Deep expertise in Apache Spark and PySpark for distributed data processing using Spark SQL, DataFrame API, RDD transformations, broadcast variables, and partition management.
- Proficient in Apache Kafka including producers, consumers, Kafka Connect, topics, partitions, consumer groups, and offset management for distributed streaming.
- Experience with NoSQL databases including MongoDB with aggregation pipelines and change streams, Cassandra for distributed databases, and Cosmos DB.
- Strong experience with Azure Synapse Analytics including dedicated SQL pools, serverless SQL pools, Synapse pipelines, and integration with Power BI for analytics workloads.
- Strong proficiency in SQL including PostgreSQL, MySQL, SQL Server with complex joins, window functions, CTEs, stored procedures, and query optimization.
- Extensive Python experience including Pandas, NumPy, custom libraries, API integration, and error handling for data engineering pipelines.
- Strong understanding of Medallion architecture (Bronze, Silver, Gold), Lambda architecture, Kappa architecture, and data mesh principles.
- Proficient in monitoring using Azure Monitor, CloudWatch, Grafana, Prometheus with custom metrics, dashboards, and alerting.
- Experience with Great Expectations for data validation, schema validation, data profiling, and automated quality checks
- Proficient in Docker, Kubernetes, Helm, Azure Kubernetes Service (AKS), and Amazon EKS for containerization and orchestration.
- Experience with Azure Key Vault, AWS Secrets Manager, encryption, IAM, RBAC, and OAuth 2.0 for security implementation
- Proficient in data serialization formats including Parquet, Avro, ORC for columnar storage optimization, JSON for semi-structured data, CSV for delimited files.
- Proficient in CI/CD pipelines, GitLab CI/CD, GitHub Actions, and Azure Pipelines for automated testing, building, and deployment of data engineering solutions.
- Experience with Infrastructure as Code using Terraform, CloudFormation, ARM Templates for multi-cloud infrastructure provisioning.
- Experience with data serialization and deserialization using Apache Avro schema registry, schema evolution patterns, and backward/forward compatibility for streaming applications.
- Knowledge of Hadoop ecosystem including HDFS, YARN, Hive, and Sqoop for big data processing.

Technical Skills:

Cloud Platforms & Services: Azure - Azure Data Factory (ADF), Azure Databricks, Azure Data Lake Storage Gen2 (ADLS), Azure Cosmos DB, Azure Event Hubs, Azure Functions, Azure Fabric, Azure DevOps, Azure Key Vault**, AWS** - AWS Glue, Amazon EMR, Amazon S3, AWS Lambda, AWS Step Functions, Amazon Kinesis, AWS Athena, AWS CloudWatch, AWS CloudFormation**, GCP** - Cloud Storage, Pub/Sub, Dataflow, cloud composer
Database and Warehousing
PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Cassandra, Cosmos DB, DynamoDB, Azure Synapse Analytics, Amazon Redshift, Google BigQuery, Snowflake
Programming & Databases: Python, PySpark, Scala, Shell Scripting, PowerShell, MySQL, PostgreSQL, SQL Server, T-SQL
Big Data & Processing: Apache Spark, PySpark, Spark SQL, Spark Structured Streaming, Kafka, Airflow, Iceberg
Data Architecture: Lakehouse, Medallion (Bronze/Silver/Gold), Lambda Architecture, Data Mesh, Star Schema, Snowflake Schema, Data Vault, SCD Types, ETL/ELT Patterns
Orchestration & DataOps: Apache Airflow (DAGs, Custom Operators, Sensors), Azure Data Factory, AWS Step Functions, Glue Workflows, CI/CD, DevOps, Git, GitHub
Monitoring & Dashboards: Azure Monitor, CloudWatch, Tableau, Power BI, Looker
Performance & Security: Query Optimization, Partitioning, Z-Ordering, Broadcast Joins, Key Vault, Secrets Manager

Professional Experience

Senior Data Engineer — Mar 2024 – Present
AbbVie

- Led the migration from legacy ETL processes to modern Lakehouse architecture on Azure coordinating data engineering and DevOps teams, resulting in 50% faster data delivery and improved data quality.
- Developed scalable ELT pipelines using Azure Data Factory (ADF) to orchestrate daily ingestion of 10+ GB high-frequency time-series data from two distinct API sources into bronze layer of Azure Data Lake Storage Gen2.
- Built and optimized data transformation pipelines using PySpark and Spark SQL in Azure Databricks processing 100M+ records monthly, reducing processing time by 35% and enabling same-day data availability.
- Designed dimensional data model in Azure Synapse Analytics with optimized fact and dimension tables using Star Schema, reducing average query response times by 75% for downstream BI tools.
- Developed compliant data de-identification service using custom Python hashing logic in Databricks, generating consistent anonymized identifiers across 5+ source systems for secure analytics while maintaining regulatory compliance.
- Designed dimensional data model in Azure Synapse Analytics with optimized fact and dimension tables using Star Schema, reducing average query response times by 75% for downstream BI tools.
- Implemented Delta Lake format in Databricks enabling zero-downtime schema evolution with ACID transactions, supporting 10+ schema changes over 6 months while maintaining continuous data availability.
- Configured comprehensive monitoring and alerting using Azure Data Factory and Azure Monitor with automated failure notifications and retry logic, reducing pipeline failures by 60% and incident response time from 4 hours to 30 minutes.
- Developed PySpark integration pipeline to merge streaming and batch data from 5+ sources using DataFrame API and Spark SQL, creating unified datasets for cross-source analysis and reducing manual data preparation by 40 hours monthly.
- Collaborated with cross-functional teams including product owners and architects to translate business requirements into technical solutions and define integration requirements for scalable data platform implementation.
- Developed Azure Functions in Python to invoke custom Python scripts for data transformations and analytics on large datasets in Databricks clusters with event-driven triggers.
- Implemented serverless architecture using Azure API Management, Azure Functions, and Cosmos DB to trigger Azure Functions for processing incoming data in Blob Storage and process pipeline jobs.
- Created analytics dashboards and reports using Azure Synapse Analytics and Power BI to provide actionable insights into operational efficiency and key business performance metrics for executive stakeholders.
- Configured and managed Azure services including Azure SQL Database, Blob Storage, Azure Event Hubs, Azure Queue Storage for building event-driven data ingestion pipelines.
- Optimized Spark cluster configurations and Delta Lake table structures using partition pruning, Z-ordering, and broadcast joins, reducing query latency and improving performance for downstream analytics.
- Built incremental data loading strategies using CDC patterns in Azure Data Factory and PySpark with timestamp-based watermarking and upsert operations for efficient data synchronization.

Senior Data Engineer — Aug 2022 – Mar 2024
Disney

- Engineered PySpark Structured Streaming pipelines on Azure Databricks processing 1B+ daily playback events using Spark SQL and DataFrame API to calculate viewing duration metrics for downstream monetization and billing systems.
- Optimized Spark cluster configurations and Delta Lake table structures using Z-ordering, partition pruning, and broadcast joins, reducing pipeline processing latency by 45% and enabling faster metric availability for downstream analytics.
- Developed data validation framework using Python and PySpark enforcing schema compliance on raw JSON events, improving data quality by identifying 75% more invalid records.
- Designed Star Schema data model in Azure Synapse Analytics with optimized fact and dimension tables serving 50+ business users for daily revenue reporting and analytics.
- Developed ETL solutions for Business Intelligence platforms using Azure Data Factory, Databricks, and Synapse Analytics.
- Installed and configured Apache Kafka clusters with producers and consumers in Python and Java for distributed streaming connections from source systems to Azure.
- Implemented Apache Airflow to orchestrate 20+ interdependent pipelines across Databricks and Data Factory, developing custom operators and dynamic DAGs with SLA monitoring that reduced manual coordination by 65%.
- Created Azure Functions in Python to receive events from Blob Storage, implementing event-driven triggers for automated data ingestion.
- Identified performance bottlenecks in Spark jobs, analyzed execution plans, and implemented optimization techniques including catalyst optimizer tuning.
- Developed PySpark jobs using DataFrame API and RDD transformations with partition management, caching strategies, and broadcast variables.
- Developed REST API integration layer using Python with OAuth 2.0 authentication, rate limiting, and retry logic for external data ingestion.

Senior Data Engineer — May 2020 – Aug 2022
Synchrony

- Engineered scalable event collection pipelines using Azure Data Factory to ingest high-volume event logs from external marketing APIs, internal mobile services, and MongoDB processing 500M+ records daily into ADLS Gen2.
- Optimized PySpark transformations using Delta Lake tables to enforce schema validation on high-volume event data, reducing join processing time by 40% through strategic data partitioning and improving downstream analytics performance.
- Engineered incremental loading strategy using CDC patterns in Azure Data Factory and PySpark to process only delta changes hourly, reducing processing overhead by 60% compared to full-table reloads.
- Configured Azure Databricks clusters with autoscaling and optimized Spark configurations for large-scale data processing workloads.
- Created Azure Functions in Python to trigger automated data pipeline workflows based on Blob Storage events and scheduled timers.
- Created T-SQL stored procedures and views in Azure SQL Database and Synapse Analytics for data transformation and reporting requirements.
- Built data pipelines using Azure Data Factory and Azure Databricks to process semi-structured JSON data from REST APIs, implementing error handling and retry logic for reliable data ingestion.
- Implemented Azure Event Hubs for real-time event streaming from mobile applications and web services, processing high-velocity clickstream data for customer behavior analytics.
- Created Azure Functions in Python to trigger automated data pipeline workflows based on Blob Storage events and scheduled timers.

Data engineer — Feb 2018 – Apr 2020
Liberty Mutual

- Streamlined automated ETL workflows using AWS Glue and Lambda functions for event-driven data ingestion from 3+ external sources, ensuring reliable weekly delivery to the marketing automation platform
- Optimized SQL query performance on Amazon Redshift by analyzing slow queries and implementing column encoding, Sort Keys, and Distribution Keys for optimized data distribution.
- Orchestrated end-to-end ETL workflows using Apache Airflow on EC2 to coordinate 20+ daily AWS Glue jobs and Redshift data loads, managing dependencies across bronze, silver, and gold layer transformations.
- Optimized weekly aggregation by 75% through S3 partitioning strategies and query optimization using predicate pushdown, while implementing CloudWatch monitoring and automated archiving.
- Integrated semi-structured JSON documents from S3 Bronze layer using PySpark's JSON parser in AWS Glue, extracting nested metadata and enriching master dimension records in the silver layer.
- Optimized SQL query performance on Amazon Redshift by analyzing slow queries and implementing column encoding, Sort Keys, and Distribution Keys for optimized data distribution.
- Configured AWS CloudWatch dashboards and alarms for monitoring Glue jobs, Redshift performance, and S3 metrics with automated SNS notifications.

Data Engineer — Oct 2016 – Feb 2018
NRG

- Migrated on-premises data to AWS S3 using Lambda-triggered pipelines, implementing full and delta load strategies for weekly marketing campaigns, ensuring 100% data integrity for downstream analytics.
- Implemented daily incremental loading using AWS Glue ETL that compared incoming snapshot files against existing Dim_Customer table, reducing processing times for unchanged data records.
- Contributed to CloudFormation templates for Infrastructure-as-Code, automating provisioning of S3 buckets and AWS Glue resources, speeding up environment setup time.
- Created SQL queries and stored procedures in Redshift for data aggregation, reporting requirements, and dimensional model maintenance.
- Developed Python and PySpark scripts in AWS Glue for data extraction, transformation, and loading from multiple source systems to Redshift data warehouse.
- Designed and implemented dimensional data models in Redshift using Star Schema with fact and dimension tables for marketing analytics workloads.
- Configured AWS CloudWatch for monitoring Glue and Lambda execution with automated SNS alerts for pipeline failures.

Certifications:

- **Databricks Certified:** Data Engineer Professional (2023)
- **Microsoft Certified:** Azure Data Engineer Associate (2020)

Education:

- Bachelor of Engineering, Anna University, 2005